1. Design-based inference rests on expectation and independence

Expectation averages over uncertainty

For a discrete random variable Y,

$$E[Y] = \sum_y y\, P(Y = y).$$

Conditioning restricts the relevant distribution:

$$E[Y \mid X = x] = \sum_y y\, P(Y = y \mid X = x).$$

In an experiment, potential outcomes can be treated as fixed; repeated random assignments generate the uncertainty in an estimator.

Unbiasedness is a repeated-assignment claim

An estimator $\widehat{\theta}$ is unbiased for θ if

$$E[\widehat{\theta}] = \theta.$$

Thus an unbiased estimator need not equal the truth in one realised experiment; it returns the truth *on average* over the assignment distribution.

Probability supplies the comparison rules

- $P(A) \geq 0$, $P(\Omega) = 1$, and probabilities add for mutually exclusive events.
- Conditional probability satisfies

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

- Independence means learning one event does not change the probability of the other:

$$A \perp\!\!\!\perp B \quad \iff \quad P(A \mid B) = P(A).$$

Conditional independence is often enough

Variables may be dependent overall but independent within covariate strata:

$$A \perp\!\!\!\perp B \mid C.$$

Causal identification commonly asks whether

$$\big(Y(1), Y(0)\big) \perp\!\!\!\perp T \mid X.$$

2. Independence turns observed group means into the ATE

Keep the estimand distinct from the estimator

For unit i,

$$\tau_i = Y_i(1) - Y_i(0).$$

Two population summaries answer different questions:

$$\mathrm{ATE} = E[Y(1) - Y(0)],$$
$$\mathrm{ATT} = E[Y(1) - Y(0) \mid T = 1].$$

The observed difference in means,

$$\widehat{\tau}_{\mathrm{DiM}} = \bar{Y}_{T=1} - \bar{Y}_{T=0},$$

is an estimator, not automatically a causal effect.

Randomisation makes the estimator unbiased

Under random assignment,

$$\big(Y(1), Y(0)\big) \perp\!\!\!\perp T,$$

so treated and control units represent the same potential-outcome distributions. Consequently,

$$E[\widehat{\tau}_{\mathrm{DiM}}] = \mathrm{ATE}.$$

Without independence, the contrast mixes two quantities

Add and subtract the treated group's missing control mean:

$$\begin{aligned}
&E[Y(1) \mid T = 1] - E[Y(0) \mid T = 0] \\
&= \underbrace{E[Y(1) - Y(0) \mid T = 1]}_{\mathrm{ATT}} \\
&\quad + \underbrace{E[Y(0) \mid T = 1] - E[Y(0) \mid T = 0]}_{\text{selection bias}}.
\end{aligned}$$

The observed contrast is a *prima facie* effect: it becomes causal only when the assignment mechanism justifies the required exchangeability.

Core warning: $E[Y(1)]$ and $E[Y(1) \mid T = 1]$ need not be equal. The people who receive treatment may differ systematically in how they would fare under either condition.

Design question: what made some units treated and others controls?

3. A credible experiment defines both the treatment and its assignment

Define an intervention that can vary

- Fix treatment timing; covariates used for adjustment must be causally prior to treatment.
- Specify the treatment and control conditions closely enough that the potential outcomes are meaningful.
- Be cautious with immutable attributes: "race," for example, bundles many social signals and mechanisms.

Attribute experiments identify manipulable signals

In the Bertrand–Mullainathan résumé experiment, randomly assigned names signalled perceived race while qualifications were held fixed. The design identifies the effect of that signal on callbacks—not a single, context-free "effect of race."

Interpretation rule: name the intervention actually randomised, including its timing and version.

Choose an assignment mechanism

- **Bernoulli:** assign each unit independently with probability π; the treated count is random.
- **Complete randomisation:** fix the number or proportion assigned to treatment.
- **Blocked randomisation:** randomise within pre-treatment groups to improve balance.
- **Cluster randomisation:** assign groups rather than individuals when delivery or spillovers operate at a higher level.
- Waitlists and stepped wedges vary when treatment becomes available.

Assignment selects the observed outcome

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

The mechanism does not create the potential outcomes; it determines which one is observed and which becomes missing.

Key idea: an assignment mechanism is a missing-data-generating procedure.

4. The assignment rule can reverse the sign of the observed contrast

Ignorability describes what assignment can depend on

An assignment mechanism is ignorable when missing potential outcomes add no information about treatment once observed data and covariates are known:

$$P(T \mid X, Y_{\mathrm{obs}}, Y_{\mathrm{mis}}) = P(T \mid X, Y_{\mathrm{obs}}).$$

The stronger unconfoundedness condition is

$$P(T \mid X, Y_{\mathrm{obs}}, Y_{\mathrm{mis}}) = P(T \mid X),$$

equivalently,

$$\big(Y(1), Y(0)\big) \perp\!\!\!\perp T \mid X.$$

Randomisation achieves this by design; observational studies must defend it using substantive knowledge and measured pre-treatment covariates.

If assignment uses unobserved potential outcomes, the observed groups are deliberately incomparable.

The "perfect doctor" treats those who benefit

Patient	$Y(0)$	$Y(1)$	τ_i	T	Y_{obs}
1	1	6	5	1	6
2	3	12	9	1	12
3	9	8	-1	0	9
4	11	10	-1	0	11
Mean	6	9	3		

The observed comparison says treatment is harmful:

$$\bar{Y}_{T=1} - \bar{Y}_{T=0} = 9 - 10 = -1.$$

Yet the full schedule shows a beneficial average effect:

$$\mathrm{ATE} = 9 - 6 = 3.$$

The doctor assigns treatment using who would benefit, so assignment is non-ignorable and confounded.

Conclusion: even perfect targeting can make a naïve treatment–control comparison badly misleading.